developments: the intellectual movement known as the Enlightenment (c. 1715–1800), and the Industrial Revolution (1760–1840).

The Enlightenment was marked by scientific progress and increased questioning of religious orthodoxy, such as the claim that God created Earth and all living things in seven days. Then, as the Industrial Revolution gathered pace, canals, railroads, mines, and quarries cut through rock strata and revealed thousands of fossils, mostly of animal and plant species that no longer existed and had never been seen before. These suggested that life began long before the widely accepted creation date of 4400 BCE, deduced from biblical sources.

Animal adaptation

In the late 1700s, French scientist Georges-Louis Leclerc, Comte de Buffon, upset church authorities by asserting that Earth was much older than the Bible suggested. He believed it was formed from molten material, struck off the Sun by a comet, that had taken 70,000 years to cool (a huge underestimate, in fact). As Earth cooled, species had appeared, died off, and were finally replaced by ancestors of those known today. Noting similarities among animals such as lions, tigers, and cats, Buffon deduced that 200 species of quadrupeds had evolved from just 38 ancestors. He also believed that changes in body shape and size in related species had occurred in response to living in different environments.

In 1800, French naturalist Jean-Baptiste Lamarck went further. In a lecture at the Museum of Natural History in Paris, he argued that traits acquired by a creature during its lifetime could be inherited by its offspring—and that a buildup of such changes over many generations could radically alter an animal’s anatomy.

Lamarck wrote several books in which he developed this idea of transmutation. He argued, for instance, that the use or nonuse of body parts